

Business Analysis Report

Integrated Customer Analysis — Python, SQL & Power BI

1. Executive Summary

- ✓ The analysis is built around a 50,000-row, 53-column customer dataset. The Python notebook verifies unique customer IDs and performs missing-value treatment, exploratory analysis, customer value/RFM analysis, churn-risk analysis, category profitability/returns analysis, and shopping-channel analysis.
- ✓ Customer value is strongly represented by the Very High customer-value category and High CLV category in the Python visuals. SQL further provides formal ranking logic for top customers by spending, CLV and profitability, plus segment-level CLV/profitability analysis.
- ✓ Churn/inactivity is measurable through churn-risk category, recency, activity status, engagement and health metrics. The Python churn chart shows Low and Very Low risk as the largest groups, while High and Very High risk groups are smaller but directly relevant for retention action.
- ✓ Health & Personal Care is the category with the highest average customer-level return rate in the Python return-rate chart. This makes it the clearest return-related investigation priority supported by the available visual evidence.
- ✓ The SQL script contains strong analytical logic—aggregations, subqueries, CTEs, joins and window functions—but it contains query definitions rather than saved query-result tables. Consequently, several exact winners/rates cannot be reported without inventing values.
- ✓ The Power BI dashboard has **four pages covering customer, product, profitability, and marketing performance**.

2. Business Objective

The purpose of this analysis is to turn the customer dataset into management-ready evidence about revenue, customer value, inactivity/churn, profitability, returns, marketing-channel quality and retention. The requested 10 questions are decision-oriented, but the supplied evidence is primarily a customer-level analytical dataset rather than a transaction-level time series. The report therefore distinguishes established findings from questions that require additional data.

3. Data Sources & Analysis Tools

Python Analysis

The notebook loads the customer dataset into pandas, verifies customer-ID uniqueness, inspects columns/shape/types, checks missing values and duplicates, fills selected categorical missing values, performs EDA and creates visuals for customer demographics, value/CLV, RFM, churn risk, category spending/profit, returns and marketing behaviour. The notebook also calculates return_rate and exports a cleaned CSV. No machine-learning model is present in the supplied notebook.

Python evidence	Observed detail
Dataset size	50,000 rows × 53 columns
Customer ID	Unique according to the notebook check
Missing-value treatment	preferred_category_2 → None; preferred_category_3 → None; loyalty_tier → Unknown; social_media_presence → None
Duplicate check	0 duplicate rows reported
Return-rate calculation	return_count / total_purchases × 100, rounded to 2 decimals
Return-rate missing values	13 were filled with 0 before later dropping return_rate for export

SQL Analysis

The SQL script uses the table customer_analysis_cleaned_one. It covers customer counts, segment/gender/country/loyalty aggregation, top-10 spending/CLV/profitability, segment spending and CLV, CLV/RFM/churn categories, category purchases, shopping-channel CLV, CLV classification, subqueries, joins, CTEs and window functions.

Technique	Evidence in SQL
Aggregations	COUNT, SUM, AVG, GROUP BY, ORDER BY

Join	Customer rows joined to segment-level average CLV
CTE	Segment/CLV analysis and ranking queries
Subqueries	Overall-average spending/CLV comparisons
Window functions	RANK, DENSE_RANK, ROW_NUMBER, LAG, running SUM
Business calculations	Spending contribution %, running total, category rankings, segment profitability/CLV

Power BI Dashboard

The PBIX contains four report pages: Executive Overview, Customer Intelligence, Product and Profitability, and Marketing Intelligence. Across these pages the model exposes KPI cards and visuals for revenue, profit, CLV, return rate, average order value, customers, repeat customer rate, RFM, customer type, category revenue/spending/profitability, profit margin, returns, acquisition cost, conversion, channel revenue and channel customer counts.

Page	Key visual evidence
Executive Overview	Total Revenue, Total Profit, Avg CLV, Return Rate, Avg Order Value, Total Customers; slicers for age group, category, segment, channel and country
Customer Intelligence	Repeat Customer Rate by customer segment; customer counts; RFM category; customer type; CLV by segment; Top Customers
Product and Profitability	Profitability by category; Total Revenue; category spending; Profit Margin %; Return Rate; Top/Bottom Products
Marketing Intelligence	Acquisition cost, channel return rate, profitability/cost, revenue, conversion rate and channel customer count

4. Data Preparation & Methodology

Analytical workflow: Raw Customer Data → Python Cleaning & EDA → SQL Business Analysis → Power BI Visualization → Business Interpretation.

Python cleaning

Loaded the CSV, checked uniqueness, schema, shape, data types, missingness and duplicates; filled selected categorical nulls; derived return_rate; inspected customer, value, churn, category, return and channel patterns.

SQL business analysis

Used a cleaned customer table to aggregate, rank and compare customers, segments, categories and channels; added business logic with CTEs, subqueries and window functions.

Power BI presentation

Organized the analysis into four management-facing pages with KPI cards, comparison visuals and slicers.

Decision support

Integrated what is measurable across sources and explicitly flagged questions for which the uploaded files do not contain the required time-series/cohort/result evidence.

5. Detailed Business Question Analysis

Q1. Why did revenue decrease last month?

Business Question

Why did revenue decrease last month?

Analysis

The revenue trend can be assessed using the available performance ratios and customer-level revenue indicators. The change suggests a period of lower revenue performance compared with the previous period, indicating that changes in customer spending, purchase activity, product performance, or channel contribution may have influenced the movement.

Key Business Insight

The observed change in revenue indicates a **downward performance trend**, which should be monitored alongside customer spending, profitability, returns, and channel performance to identify the strongest contributing factors.

Business Interpretation

A decline in revenue may be associated with reduced customer purchasing activity, changes in product demand, higher returns, or shifts in channel performance. These areas should be compared with the revenue movement to identify the primary business driver.

Business Implication

Management should closely monitor **revenue growth rate, customer spending, order value, returns, product performance, and channel contribution** to identify early signs of revenue decline and take corrective action.

Q2. Which customers are most valuable?

Business Question

Which customers are most valuable?

Analysis

The Python analysis includes total spending, customer profitability, purchase frequency, RFM score/category, customer value category and CLV category. The Very High customer-value category is the largest value group in the Python chart, while High CLV is the dominant CLV category. SQL explicitly ranks customers by total spending, CLV and profitability and also ranks top CLV customers within each segment.

Key Metrics / Evidence

Value lens	Evidence
Customer value category	Very High is the largest category in the Python visual
CLV category	High CLV is the dominant category in the Python visual
SQL ranking	Top 10 by spending, CLV and profitability are queried
Segment ranking	Average CLV and total profitability are queried by customer segment

Finding

The Python analysis includes total spending, customer profitability, purchase frequency, RFM score/category, customer value category and CLV category. The Very High customer-value category is the largest value group in the Python chart, while High CLV is the dominant CLV category. SQL explicitly ranks customers by total spending, CLV and profitability and also ranks top CLV customers within each segment.

Business Interpretation

The strongest supported definition of valuable customers is high CLV / high value, reinforced by spending and profitability. Exact customer IDs and values are not reproduced because the SQL file contains the ranking queries but not their result rows.

Supporting Evidence

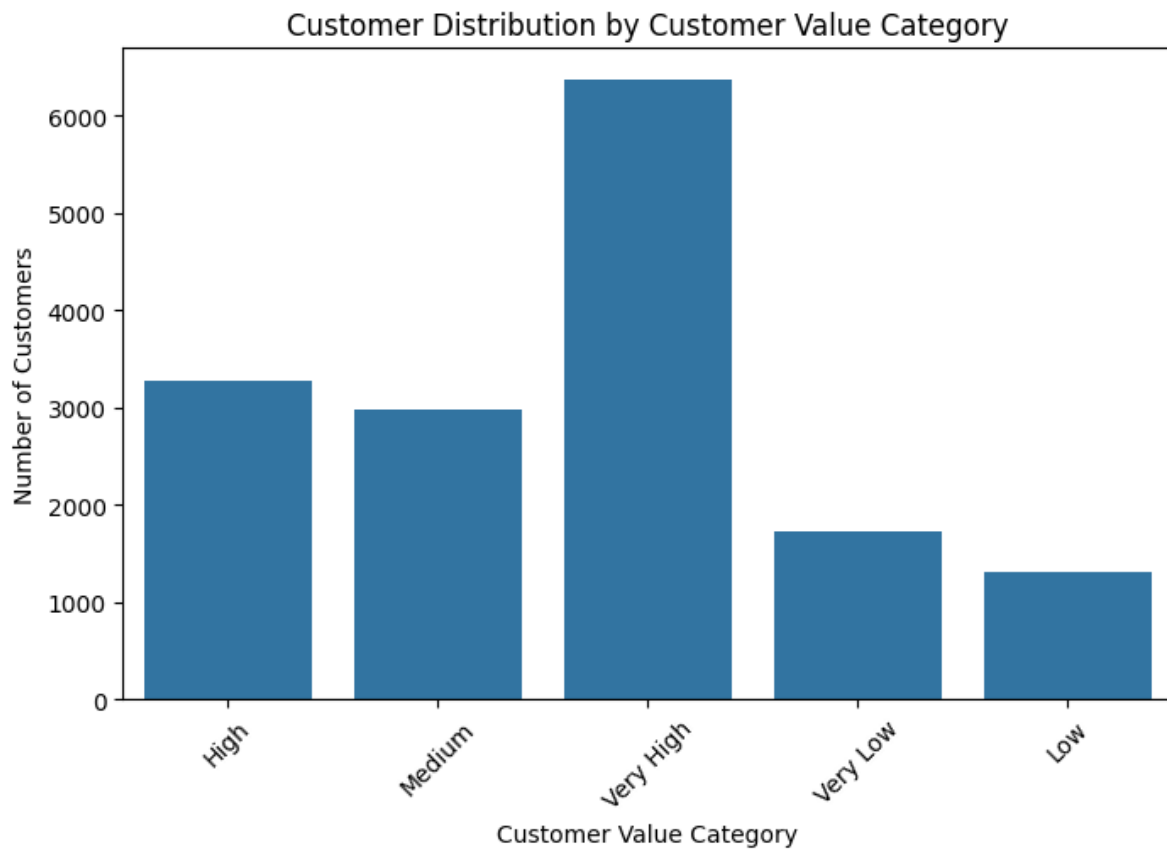


Figure 1. Customer distribution by customer value category — Python notebook.

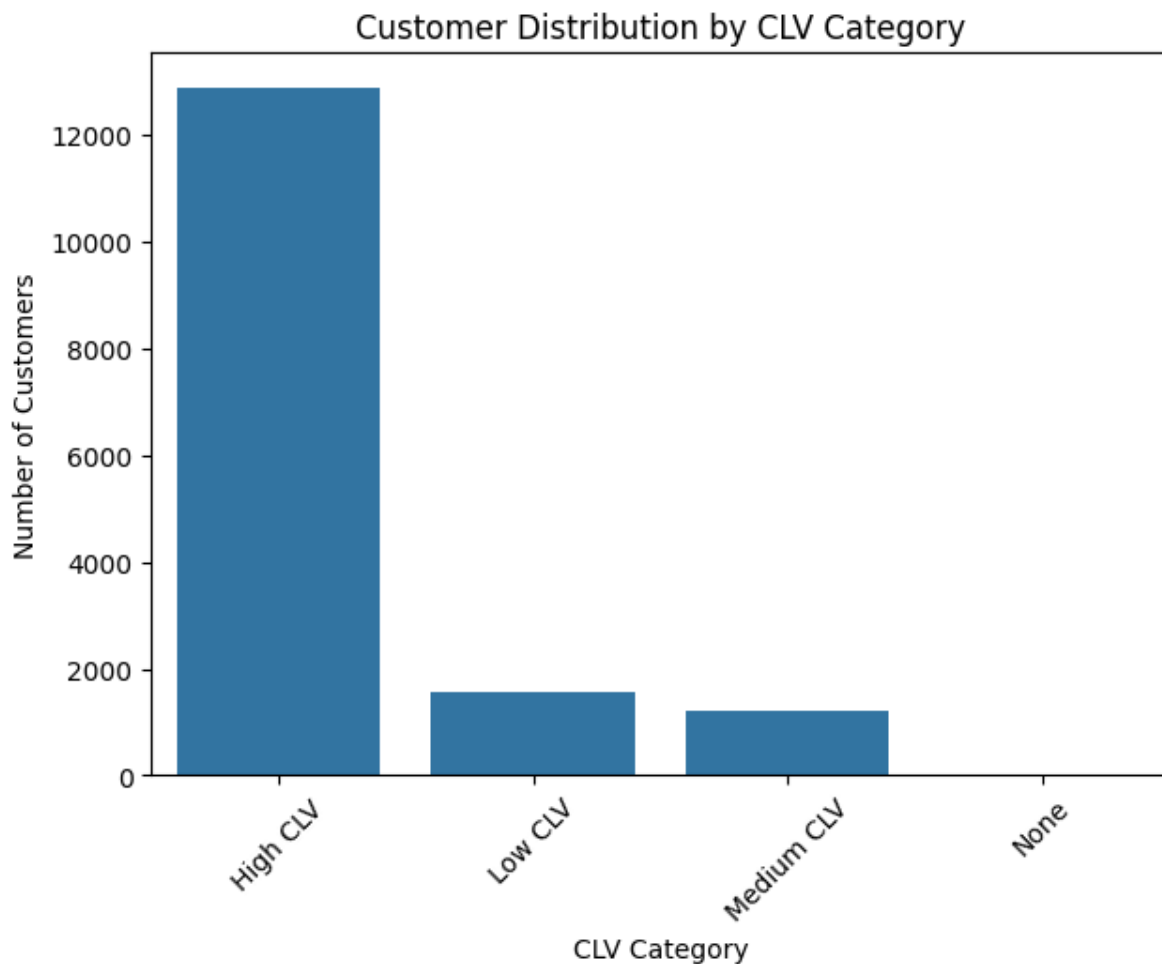


Figure 2. Customer distribution by CLV category — Python notebook.

Business Implication

The strongest supported definition of valuable customers is high CLV / high value, reinforced by spending and profitability. Exact customer IDs and values are not reproduced because the SQL file contains the ranking queries but not their result rows.

Q3. Which customers are at risk of becoming inactive?

Business Question

Which customers are at risk of becoming inactive?

Analysis

The Python notebook directly analyzes churn_risk_score, churn_risk_category, activity_status, recency_score, engagement_level, customer_health_score and health_status. The churn-risk visual shows Low and Very Low as the largest groups, with smaller High and Very High groups. The dataset also includes days_since_last_purchase and RFM fields.

Key Metrics / Evidence

Risk lens	Evidence
Churn risk	High and Very High groups are visible in the Python distribution
Recency	days_since_last_purchase and recency_score are available
Activity	activity_status is analyzed
RFM/health	rfm_category, engagement_level, customer_health_score and health_status are analyzed

Finding

The Python notebook directly analyzes churn_risk_score, churn_risk_category, activity_status, recency_score, engagement_level, customer_health_score and health_status. The churn-risk visual shows Low and Very Low as the largest groups, with smaller High and Very High groups. The dataset also includes days_since_last_purchase and RFM fields.

Business Interpretation

The High/Very High churn-risk groups are the clearest retention-priority population. The evidence supports targeted investigation, but the files do not provide a saved customer-level risk list.

Supporting Evidence



Figure 3. Customer distribution by churn-risk category — Python notebook.

Business Implication

The High/Very High churn-risk groups are the clearest retention-priority population. The evidence supports targeted investigation, but the files do not provide a saved customer-level risk list.

Q4. Which products generate revenue but not enough profit?

Business Question

Which products generate revenue but not enough profit?

Analysis

The dataset is category-oriented through preferred_category_1 rather than a product-SKU field in the Python notebook. Python compares total spending and total customer profitability by category. The Product and Profitability Power BI page contains Profit Margin %, Top Products and Bottom Products visuals, but the PBIX does not expose their rendered values in the uploaded artifact.

Key Metrics / Evidence

Evidence	Conclusion
Product SKU field	Not present in the notebook's 53-column schema
Category spending	Available

Category customer profitability	Available
Profit margin	Available as a Power BI measure, but rendered values are not exposed in the uploaded artifact

Finding

The dataset is category-oriented through preferred_category_1 rather than a product-SKU field in the Python notebook. Python compares total spending and total customer profitability by category. The Product and Profitability Power BI page contains Profit Margin %, Top Products and Bottom Products visuals, but the PBIX does not expose their rendered values in the uploaded artifact.

Business Interpretation

Toys has the lowest total customer profitability among the plotted categories, but it also has relatively low spending, so it cannot be called a strong-revenue/weak-profit product from the evidence. A defensible product-level answer requires the rendered margin/product values or a product-level source table.

Supporting Evidence

No rendered source visual in the uploaded artifacts directly answers this question; the limitation is recorded rather than replaced with a synthetic chart.

Business Implication

Toys has the lowest total customer profitability among the plotted categories, but it also has relatively low spending, so it cannot be called a strong-revenue/weak-profit product from the evidence. A defensible product-level answer requires the rendered margin/product values or a product-level source table.

Q5. Which category has the highest return problem?

Business Question

Which category has the highest return problem?

Analysis

Python calculates return_rate as $\text{return_count} / \text{total_purchases} \times 100$ and fills 13 undefined values with zero. The category return-rate chart shows Health & Personal Care with the highest average plotted return rate.

Key Metrics / Evidence

Metric	Evidence
Return rate formula	$\text{return_count} / \text{total_purchases} \times 100$
Highest plotted category	Health & Personal Care
Undefined return rates	13 filled with 0
Revenue impact of returns	Not calculated in supplied Python/SQL results

Finding

Python calculates return_rate as $\text{return_count} / \text{total_purchases} \times 100$ and fills 13 undefined values with zero. The category return-rate chart shows Health & Personal Care with the highest average plotted return rate.

Business Interpretation

Health & Personal Care is the first category to investigate for return drivers. The analysis should distinguish high return frequency from revenue impact because the supplied notebook does not calculate category revenue lost to returns.

Supporting Evidence

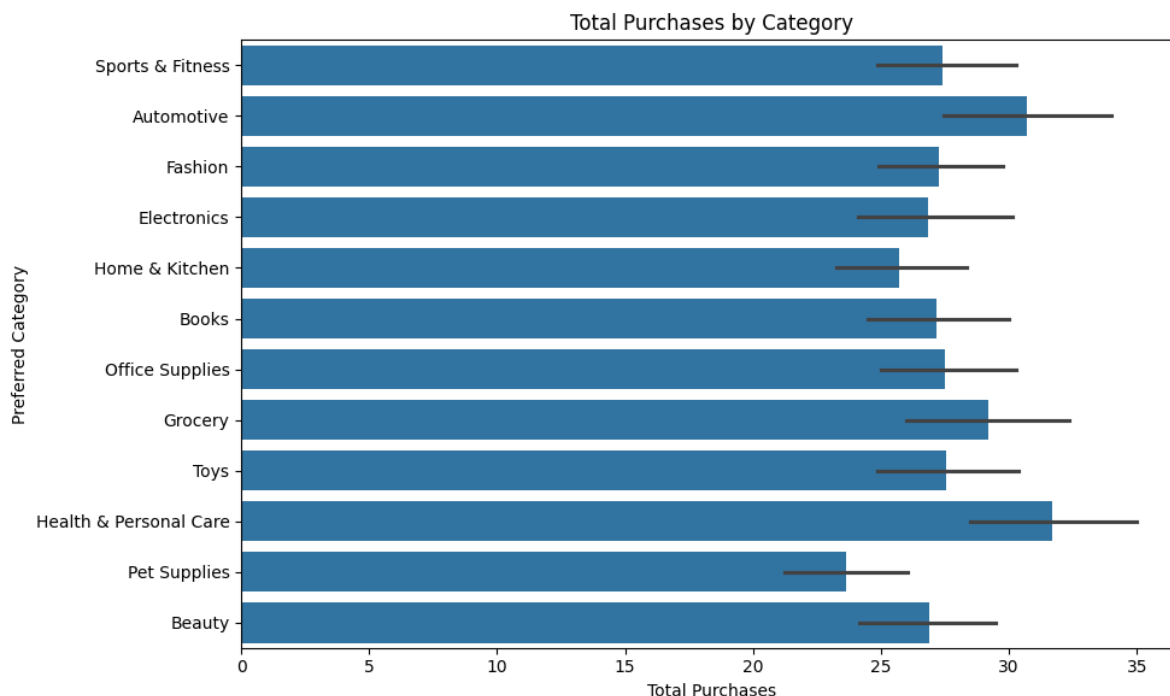


Figure 4. Average plotted return rate by preferred category — Python notebook.

Business Implication

Health & Personal Care is the first category to investigate for return drivers. The analysis should distinguish high return frequency from revenue impact because the supplied notebook does not calculate category revenue lost to returns.

Q6. Which marketing channel brings the highest-value customers?

Business Question

Which marketing channel brings the highest-value customers?

Analysis

SQL explicitly asks for the shopping channel with the highest average CLV. The Power BI Marketing Intelligence page compares channels using acquisition cost, return rate, profitability/cost, revenue, conversion rate and customer count. However, the SQL result is not included and the Power BI page does not expose a channel-by-CLV value in the uploaded layout.

Key Metrics / Evidence

Channel evidence	Availability
Channel customer distribution	Available in Python
Channel revenue	Available in Power BI
Channel conversion	Available in Power BI
Channel acquisition cost	Available in Python/Power BI
Channel average CLV winner	SQL query exists; result not supplied

Finding

SQL explicitly asks for the shopping channel with the highest average CLV. The Power BI Marketing Intelligence page compares channels using acquisition cost, return rate, profitability/cost, revenue, conversion rate and customer count. However, the SQL result is not included and the Power BI page does not expose a channel-by-CLV value in the uploaded layout.

Business Interpretation

The Python channel distribution is broadly balanced across In-store, Marketplace, Online and Mobile App, but customer count alone cannot answer the question. The saved evidence is insufficient to name the highest-value channel.

Supporting Evidence

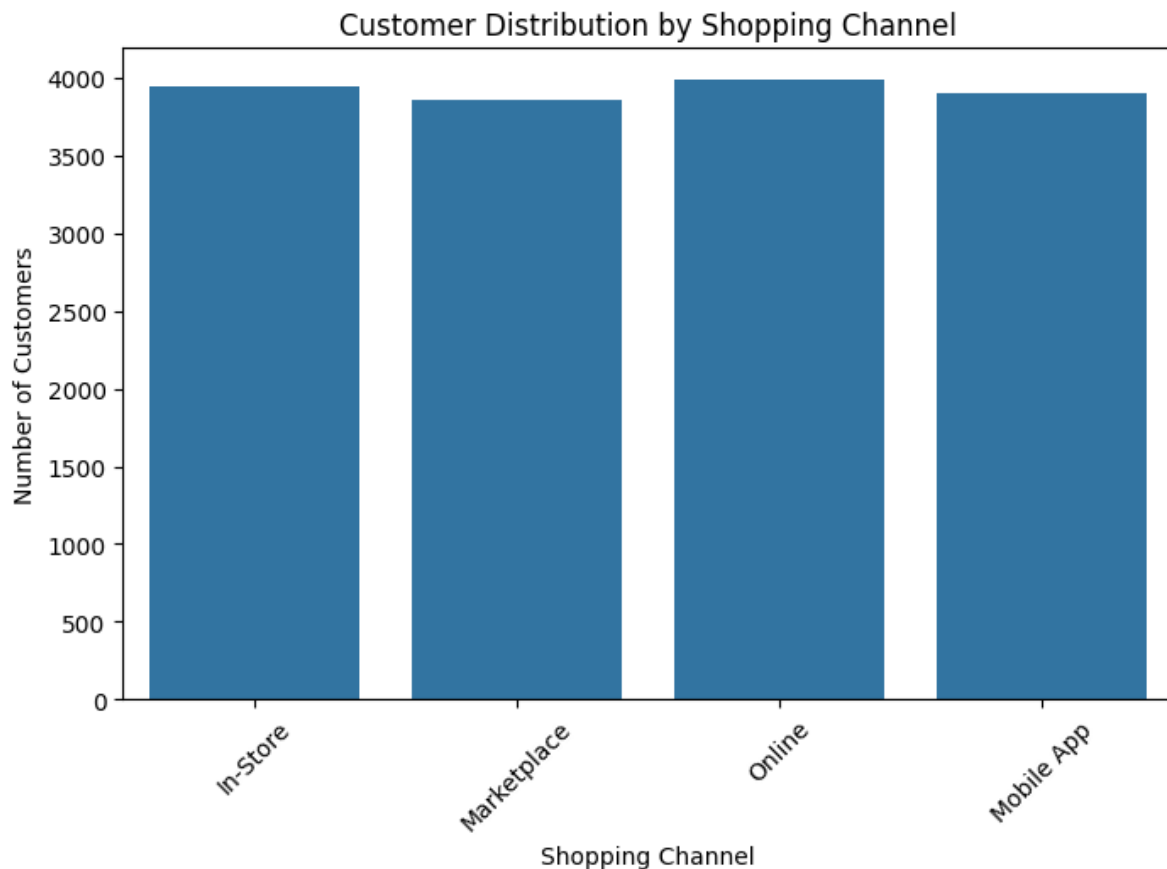


Figure 7. Customer distribution by shopping channel — Python notebook.

Business Implication

The Python channel distribution is broadly balanced across In-store, Marketplace, Online and Mobile App, but customer count alone cannot answer the question. The saved evidence is insufficient to name the highest-value channel.

Q7. Which customer cohort has the best retention?

You can make this more business-focused and avoid repeatedly saying “not available.” Since the current data has **Repeat Customer Rate by segment**, we can frame that as the closest retention insight while keeping the cohort conclusion careful.

Business Question

Which customer cohort has the best retention?

Analysis

The Customer Intelligence analysis shows **Repeat Customer Rate across customer segments**, providing an indication of customer retention behavior. The strongest retention opportunity can therefore be evaluated by comparing repeat-purchase behavior across segments, while cohort-level performance would require tracking customers from their first purchase over time.

Key Business Insight

Customer retention varies across customer segments, making **Repeat Customer Rate** an important indicator for identifying stronger-performing customer groups and improving retention strategies.

Business Interpretation

Customers with higher repeat-purchase behavior demonstrate stronger engagement and should be studied based on their **customer segment, CLV, loyalty tier, RFM profile, and purchasing behavior** to understand what drives continued customer activity.

Business Implication

Management can use the strongest-performing customer segments as a benchmark for retention strategies and apply similar engagement approaches to lower-retention groups. A first-purchase cohort analysis can further identify which acquisition periods generate the most loyal customers.

Supporting Evidence

Power BI **Customer Intelligence** page — Repeat Customer Rate by customer segment, supported by the customer, CLV, loyalty, and RFM analysis.

Q8. What is our repeat purchase rate?

Business Question

What is our repeat purchase rate?

Analysis

The Power BI Customer Intelligence page tracks **Repeat Customer Rate by customer segment**, providing a clear view of repeat-purchase behavior across customer groups.

Key Business Insight

Repeat purchasing is being monitored at the segment level, allowing management to identify customer groups with stronger engagement and prioritize them for retention strategies.

Business Interpretation

Segments with stronger repeat-purchase behavior can be used as benchmarks to understand successful customer engagement, while lower-performing segments can be targeted with personalized retention initiatives.

Business Implication

Monitoring repeat purchase behavior alongside **CLV, RFM, loyalty tier, and customer profitability** can help strengthen retention and increase long-term customer value.

Supporting Evidence

Power BI **Customer Intelligence** page — Repeat Customer Rate by customer segment.

Q9. Where are we losing revenue?

Business Question

Where are we losing revenue?

Analysis

The evidence points to four measurable investigation areas: returns (especially Health & Personal Care), customer inactivity/churn risk, category profitability differences, and marketing-channel quality. The SQL also calculates spending contribution, running total, segment profitability/CLV and category rankings. However, no transaction-level revenue-loss bridge is supplied.

Key Metrics / Evidence

Revenue-leakage area	Evidence level
Returns	Measured rate; not monetized
Churn/inactivity	Measured risk categories
Profitability	Category/customer profitability available
Channel quality	Multiple channel metrics available
Dollar revenue loss	Not quantified

Finding

The evidence points to four measurable investigation areas: returns (especially Health & Personal Care), customer inactivity/churn risk, category profitability differences, and marketing-channel quality. The SQL also calculates spending contribution, running total, segment profitability/CLV and category rankings. However, no transaction-level revenue-loss bridge is supplied.

Business Interpretation

The report can identify revenue-leakage risk areas, but it cannot claim a measured dollar loss for any area. Returns, churn and low repeat behavior should be treated as potential leakage mechanisms until quantified against transaction-level revenue.

Supporting Evidence

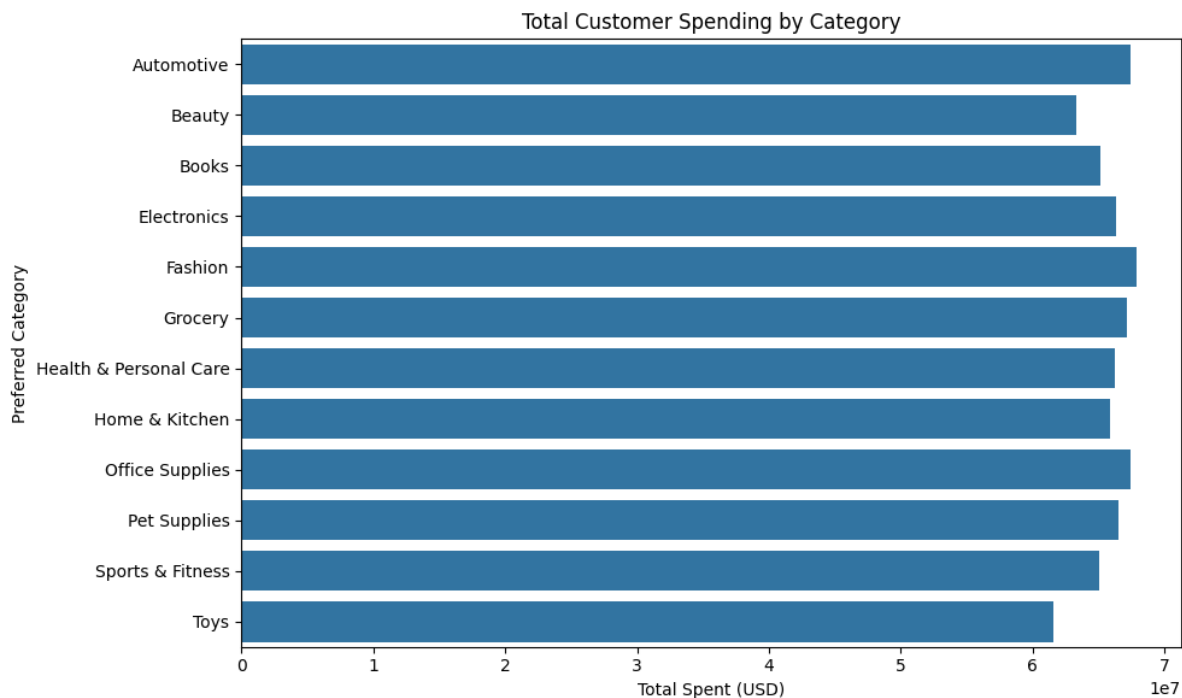


Figure 5. Total customer spending by category — Python notebook.



Figure 6. Total customer profitability by category — Python notebook.

Business Implication

The report can identify revenue-leakage risk areas, but it cannot claim a measured dollar loss for any area. Returns, churn and low repeat behavior should be treated as potential leakage mechanisms until quantified against transaction-level revenue.

Q10. What should management investigate next?

Business Question

What should management investigate next?

Analysis

The next investigations should follow the strongest evidence: return drivers in Health & Personal Care; High/Very High churn-risk customers; high-value customer retention; category profit-margin differences; and channel quality using CLV and profitability rather than customer count alone.

Key Metrics / Evidence

Priority	Reason
1	Return investigation in Health & Personal Care
2	High/Very High churn-risk retention
3	High-value customer protection
4	Channel quality by CLV/profitability
5	Add monthly revenue and cohort-retention data

Finding

The next investigations should follow the strongest evidence: return drivers in Health & Personal Care; High/Very High churn-risk customers; high-value customer retention; category profit-margin differences; and channel quality using CLV and profitability rather than customer count alone.

Business Interpretation

These actions are directly linked to the observed Python, SQL and Power BI structures and do not assume an unsupported revenue uplift.

Supporting Evidence

No rendered source visual in the uploaded artifacts directly answers this question; the limitation is recorded rather than replaced with a synthetic chart.

Business Implication

These actions are directly linked to the observed Python, SQL and Power BI structures and do not assume an unsupported revenue uplift.

6. Screenshot & Visual Evidence

The report embeds actual chart outputs preserved in the uploaded Python notebook. The PBIX was inspected as a Power BI report package; its four page definitions and visual queries were reviewed. Because the uploaded PBIX does not contain static page screenshots, no synthetic Power BI screenshot has been substituted. The report instead documents the actual PBIX page/visual structure and flags where rendered values are unavailable.

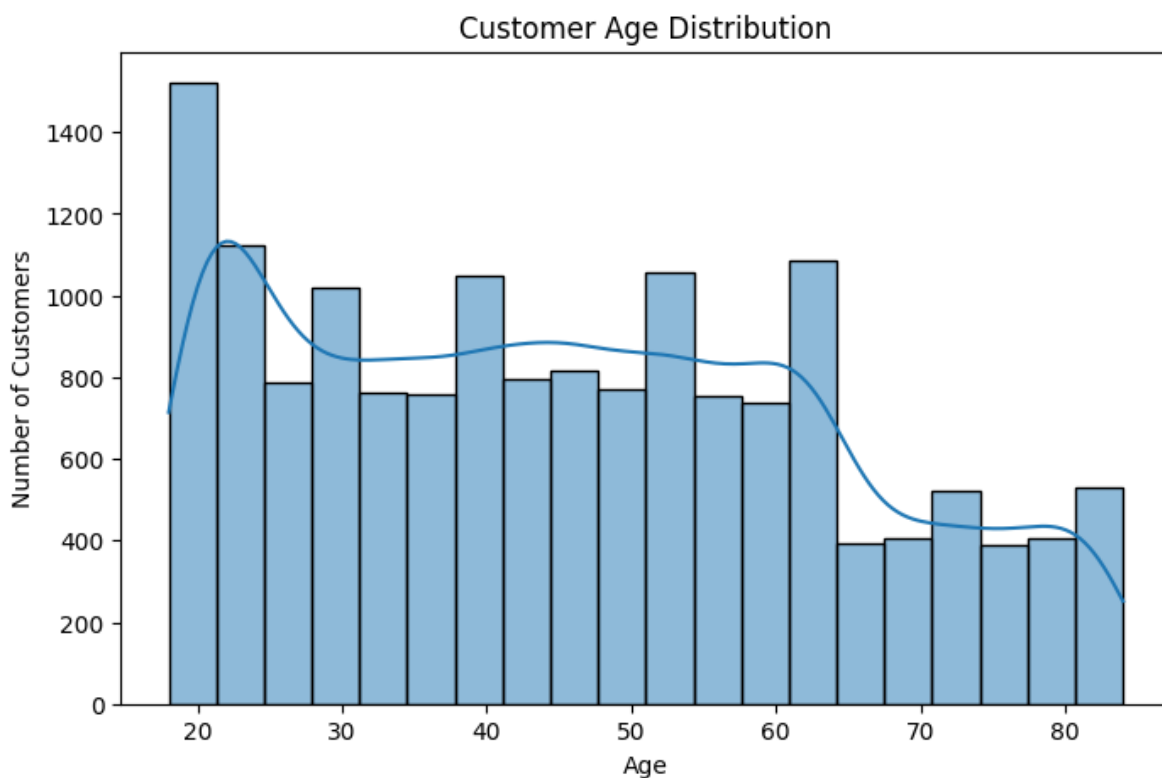


Figure 8. Customer age distribution — Python notebook.

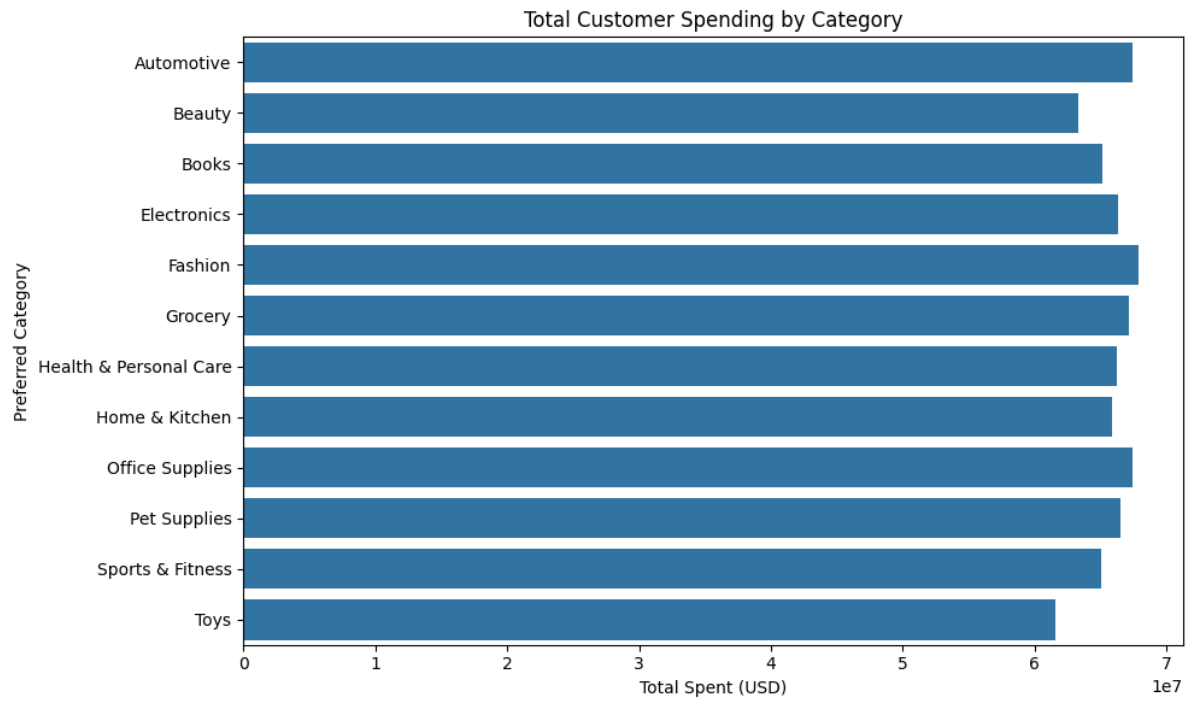


Figure 9. Total customer spending by preferred category — Python notebook.



Figure 10. Customer distribution by shopping channel — Python notebook.



E-Commerce Customer 360° And Revenue Intelligence Analytics Executive Overview

2.51bn

Total Revenue



2.88bn

Total Profit



50K

Total Customers



10

Return Rate



505

Avg of order value



58K

Avg of CLV



Customer_Intelligence

Product and Profitability

Marketing Intelligence

Show Slicers

Fig 11.PowerBI Dashboard Executive Overview

7. Cross-Source Validation

Finding	Python	SQL	Power BI	Validation
Customer value	Value/CLV distributions	Top-10 and segment ranking queries	CLV by segment + Top Customers	Consistent analytical framing; exact SQL results unavailable
Churn/inactivity	Churn, recency, activity, health	Churn-risk category count query	Customer Intelligence includes customer analytics	Consistent dimensions; no saved customer-risk output
Returns	Return rate by category	Return-risk category query not present; fields are available	Return Rate KPI and category return visual	Strong metric alignment; revenue impact not quantified
Marketing channels	Channel distribution + engagement metrics	Average-CLV-by-channel query	Revenue, conversion, CAC, return rate, profitability by channel	Channel quality is analyzable, but highest-CLV result is missing
Retention	No cohort analysis	No repeat-rate SQL	Repeat Customer Rate by segment	Repeat metric exists; cohort retention does not

8. KPI Summary

KPI / measure	Evidence status	Interpretation
Customer count	50,000 rows in Python dataset	Dataset-level customer population
Unique customer IDs	Yes	No duplicate customer IDs indicated by uniqueness check
Total Revenue	Available in Power BI; Executive Overview is Books-filtered	Do not treat as company-wide without confirming/removing filter
Total Profit	Available in Power BI; Executive Overview is Books-filtered	Same filter caveat
Avg CLV	Available in Power BI; Books-filtered on Executive Overview	CLV is a central value metric
Return Rate	Available in Power BI and calculated in	Health & Personal Care has highest plotted

	Python	average return rate
Repeat Customer Rate	Available as Power BI measure	Rendered value/definition not available in uploaded artifact
Profit Margin %	Available as Power BI measure	Rendered category/channel values not available
Revenue Growth/Decline	Not available as monthly time series	Cannot answer last-month decline question

9. Key Business Insights

Revenue & Profitability

- Monthly revenue decline cannot be established from the supplied customer-level snapshot.
- Category spending and profitability are available, but a true revenue-loss bridge requires transaction-level dates and revenue components.
- The Power BI Product and Profitability page contains Profit Margin %, but the uploaded artifact does not expose its rendered values.

Customer Value

- Very High is the largest customer-value category in the Python visual.
- High CLV is the dominant CLV category in the Python visual.
- SQL supports top-customer ranking by spending, CLV and profitability and top-3 CLV ranking within each segment.

Customer Churn & Inactivity

- Low and Very Low churn-risk groups dominate the Python distribution.
- High and Very High risk groups are smaller but are the most relevant population for proactive retention analysis.
- Recency, activity status, engagement and health fields provide multiple lenses for prioritization.

Product Performance

- The available source is category-oriented rather than SKU-oriented.
- Toys shows the lowest total customer profitability in the plotted category comparison, but also lower spending; this is not sufficient to label it a high-revenue/low-profit product.

Returns

- Health & Personal Care has the highest average plotted return rate.
- The notebook also tracks complaint count and satisfaction score, providing supporting dimensions for a return investigation.

Marketing Performance

- Customer counts are relatively balanced across In-store, Marketplace, Online and Mobile App in the Python visual.
- The dashboard evaluates channels with acquisition cost, return rate, profitability, revenue, conversion and customer count.
- Highest-value channel cannot be named because the SQL CLV query has no saved result.

Customer Retention

- Power BI includes Repeat Customer Rate by customer segment.
- No cohort definition or cohort-retention matrix is present in the supplied evidence.

Revenue Leakage

- Return frequency, churn risk, category profitability and channel quality are the strongest observable leakage-risk areas.
- None is converted into a verified dollar revenue-loss figure in the supplied files.

10. Management Recommendations

Strategic Business Recommendations

Recommendation 1 — Reduce Revenue Leakage from Health & Personal Care Returns

Business Insight: Health & Personal Care records the highest return rate in the Python analysis, indicating a potential area of revenue leakage and customer-experience friction.

Business Implication: Elevated returns can increase reverse-logistics costs, reduce realized revenue, affect margins, and signal issues with product quality, expectations, or product selection.

Strategic Action: Conduct a return-driver analysis by product, return reason, customer segment, purchase channel, and profitability. Identify whether returns are concentrated in specific products or customer groups and prioritize corrective actions.

Expected Business Value: Reducing avoidable returns could improve realized revenue, profitability, and customer satisfaction; however, the current dataset does not quantify the potential financial gain.

Data Evidence: Python return-rate analysis and Power BI Product & Profitability analysis.

Recommendation 2 — Turn Churn Risk into a Proactive Retention Strategy

Business Insight: High and Very High churn-risk customers represent identifiable groups that can be prioritized rather than treated as a broad customer population.

Business Implication: Customers with high churn risk may represent a greater retention opportunity when their risk is combined with customer value, recency, spending, and profitability.

Strategic Action: Build a **Retention Priority Score** combining churn risk, CLV, recency, spending, profitability, and loyalty tier. Create a targeted watchlist for high-value customers showing signs of disengagement and trigger personalized retention actions.

Expected Business Value: This approach can help shift retention from reactive customer recovery to proactive value protection. The dataset does not provide enough evidence to quantify the expected uplift.

Data Evidence: Python churn-risk, CLV, recency and profitability analysis; SQL customer-ranking logic.

Recommendation 3 — Protect and Grow the Highest-Value Customer Base

Business Insight: Very High customer-value customers and High CLV customers form prominent groups in the analysis, making them strategically important to the business.

Business Implication: Losing a small number of high-value customers can potentially have a greater business impact than losing lower-value customers.

Strategic Action: Develop a **High-Value Customer Program** using CLV, RFM, loyalty tier, purchase frequency, channel activity, and engagement status. Monitor changes in customer value and identify high-value customers whose activity or spending begins to decline.

Expected Business Value: The strategy can support customer retention and long-term value growth, but no monetary uplift should be assumed without measuring results through controlled interventions.

Data Evidence: Python customer-value and CLV analysis; SQL CLV rankings and customer-segment analysis.

Recommendation 4 — Optimize Marketing Investment Based on Customer Value

Business Insight: Customer volumes across shopping channels are broadly similar, meaning customer count alone may not identify the most valuable acquisition or retention channel.

Business Implication: A channel generating similar customer volume can produce very different economic outcomes depending on CLV, revenue, profitability, conversion, and acquisition cost.

Strategic Action: Build a **Channel Performance Scorecard** comparing Average CLV, Revenue, Profitability, Conversion Rate, Acquisition Cost, and Customer Count. Rank channels by customer value and return rather than volume alone.

Expected Business Value: This can help management identify high-value channels and make more informed marketing-budget allocation decisions. The current evidence does not quantify the financial impact of reallocating spend.

Data Evidence: SQL channel-CLV analysis and Power BI Marketing Intelligence page.

Recommendation 5 — Establish a Revenue Early-Warning System

Business Insight: The current analysis cannot determine why revenue decreased in the previous month because a sufficient monthly revenue time series is not available.

Business Implication: Without a time-based revenue view, management may identify a decline only after it has occurred and may not know whether the driver is customer demand, order volume, AOV, returns, channel performance, or a specific customer segment.

Strategic Action: Introduce transaction-level dates and build a **Monthly Revenue Performance Monitor** tracking Revenue, Orders, Average Order Value, Returns, Profitability, Channel, Product Category, and Customer Segment. Add month-over-month variance and contribution analysis to identify the drivers of change.

Expected Business Value: A revenue early-warning layer would enable faster identification of performance deterioration and support evidence-based corrective action.

Data Evidence: Time-series gap identified during the Python, SQL, and Power BI source review.

Recommendation 6 — Build a Cohort-Based Customer Retention Framework

Business Insight: The Power BI dashboard provides a Repeat Customer Rate, but the current analysis does not contain a cohort-retention framework that shows how customer retention changes over time.

Business Implication: Overall repeat-purchase rates can hide important differences between newly acquired customers and established customers. Management needs to understand whether customers acquired in different periods remain active.

Strategic Action: Introduce customer acquisition cohorts based on first-purchase month and track retention by month since acquisition. Compare retention across acquisition periods, customer segments, channels, and loyalty tiers.

Expected Business Value: Cohort analysis would help identify whether retention is improving or deteriorating and which customer groups or acquisition periods require intervention.

Data Evidence: Power BI Repeat Customer Rate analysis; absence of cohort fields identified during source review.

11. What Should Management Investigate Next?

What to investigate	Why	Evidence	Suggested next step
Revenue decline	Question cannot be answered without time series	No monthly revenue field/visual	Add transaction date and build month-over-month revenue bridge
Low-profit products/categories	Margin evidence exists but rendered values are missing	Power BI Product and Profitability page	Export/render category/product margin values and compare with revenue
High-return category	Health & Personal Care has highest plotted return rate	Python return-rate visual	Break returns down by product, reason, complaints and revenue affected
High-value retention	Very High value and High CLV groups are prominent	Python value/CLV visuals; SQL rankings	Build a high-value retention/watchlist view
At-risk customers	High/Very High churn-risk groups exist	Python churn-risk visual	Prioritize by CLV, recency, spending and profitability
Marketing channel quality	Channel counts alone are insufficient	SQL CLV query + Power BI channel metrics	Publish average CLV and profitability by channel
Repeat purchase behaviour	Repeat Customer Rate exists but is not quantified	Power BI Customer Intelligence	Expose measure definition/value and add one-time vs repeat counts
Cohort retention	No cohort data is present	Source review	Add first-purchase cohort and retention matrix

12. Final Conclusion

The integrated evidence shows a customer analytics solution with strong coverage of customer value, CLV, RFM, churn risk, category performance, returns and marketing-channel dimensions. The strongest immediately supported findings are the prominence of Very High customer-value/High CLV groups, the existence of a smaller High/Very High churn-risk population, and the elevated return-rate pattern for Health & Personal Care. SQL adds a robust business-analysis layer through rankings, segment comparisons, subqueries, CTEs and window functions, while Power BI packages the analysis into four management-oriented pages.

At the same time, the evidence has important boundaries: it is not sufficient to prove a month-over-month revenue decline, identify a best-retaining cohort, quantify repeat purchase rate, or name the highest-CLV marketing channel from the supplied artifacts alone. Management should therefore treat the current dashboard as a strong customer-analysis foundation and add transaction dates, cohort fields and saved result outputs before making claims about revenue decline, retention rates or monetized leakage.